



Introduction:

This is an overview of the drivetrain system with a special focus on differential mounts. These mounts are designed to firmly secure the differential at the rear end of the car in its intended location while also providing enough flexibility and movement to easily tension the chain. A key design goal is to ensure the mounts withstand the maximum amount of load that can be transferred through the drivetrain system. Secondary objectives include manufacturability, ease of access, serviceability, and weight reduction.

Goals:

1. Support the 160 mm Drexler Differential for MRX228
2. Withstand loads from the motor
3. Allow the differential movement to allow tensioning of the chain
4. Optimize the stiffness-weight ratio of the drivetrain system
5. Accommodate a 24-tooth sprocket
6. Act as rear impact protection

Rules:

F.11.2.2 Rear Impact Protection:

d. Differential mounts or a plate behind upper and lower rear impact tubes may be used as Rear Impact Protection.

If used, the mounts or plate must be 25 mm or more larger at both the top and bottom of the upper and lower tubes

F.11.2.3 All Non Crushable Items should have a minimum 25 mm clearance to the surface of the Accumulator Container.

Non Crushable Items include, but are not limited to motors, differentials, and the side or rear impact structure itself. Accumulator mounts do not require clearance

F.11.2.4 Non Crushable Items mounted behind the Rear Impact structure must not be able to come through the Rear Impact structure.

Abstract:

With the team switching from internal combustion to electric vehicle, there would be also a change in the differential mounts. NFR22 had an assembly of three parts: the differential mount, the differential holder, and the turnbuckles. The whole differential mount assembly was designed to mount onto the engine of the car, support the differential, accommodate the torque from the engine, allow for chain tensioning, and act as rear impact protection.

With NFR23, the differential mount assembly only includes two parts: the differential mount and the turnbuckles. The NFR23 differential mounts were designed to mount the differential from the frame, accommodate the torque from the motor and allow for chain tensioning.

The major challenges that were encountered were mainly understanding the different changes that came with the car switching from IC to EV. One of the team members was a freshman with zero design experience in mechanical components and therefore struggled a lot to understand the functions of the differential mounts. The other team member was a returning member with some experience which helped out a lot. However, it took a long time for them to fully understand the functions of the differential mount. Even so, more difficulty came from having to make calculations for the loads that the differential mounts had to support. Neither team member had taken relevant classes that provided knowledge on some calculations that had to be made. Luckily there was a lot of support from Drew (Chassis lead), Jake (Chief Engineer), Isabella (Suspension lead), and Benji (Sus team member). One last major challenge was also designing a component that could be manufacturable in-house. Neither team member had sufficient experience in CAM and the CNC mill to be able to fully manufacture the components without any external help. However, the suspension team came in clutch with Aidan and helped with the CAM and CNC mill operations.

The differential mounts are a drive-critical component and the team members acknowledge that a lot of knowledge and skills were needed in order to make a design that fulfilled all its requirements. Particularly, knowledge and skills in force and torque calculations, mechanical design, and Computer-Aided Manufacturing. Future steps that the team would have taken would be to improve the manufacturing process for the differential mounts, especially having to assemble the differential mounts with the bearings and the differentials. This process was not well refined and a lot of improvisation had to be done.

Design Overview:

In NFR22, the differential mount had 3 components each for the right and left differential mounts. There was a mount, a holder, and a turnbuckle. The holders actually held the differential, while the mounts held the holders. The mounts were (top) connected to the engine and frame (bottom).

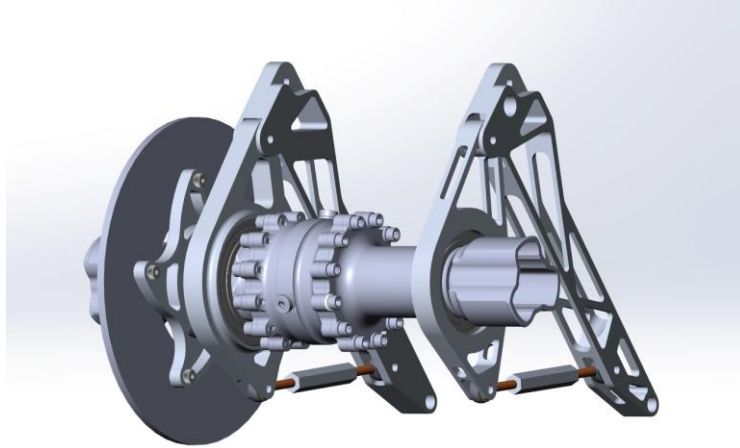


Figure 1: NFR22 Assembly

The differential mounts were designed this way because the engine was large and strong enough to withstand the forces going through the mounts. However, this year, after replacing the engine with a smaller, lighter motor, we have to connect the top mount to the frame itself. Therefore, the distance between the top mounting location and the bottom mounting location is significantly less than what it was last year.

For NFR23, the team started designing the differential holders. Throughout the eight months of the design process, there was a total of 3 design concepts that were looked at. The first design concept was to try to mimic NFR22 but just switch the mounting positioning from the engine to the frame at the back. This proved to not be able to work because NFR23 mounts have a different load case from the NFR22 therefore making a mount similar to NFR22 would not work at all. From Figure 1, the turnbuckles were at the bottom which was discussed as a change that could be made for NFR23 by moving the turnbuckles to the top and extending the mounts to the bottom bar of the frame (as shown in figure 2).

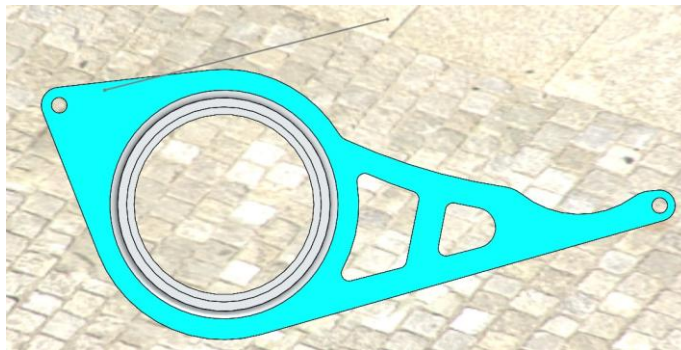


Figure 2: Right Differential Mount #2

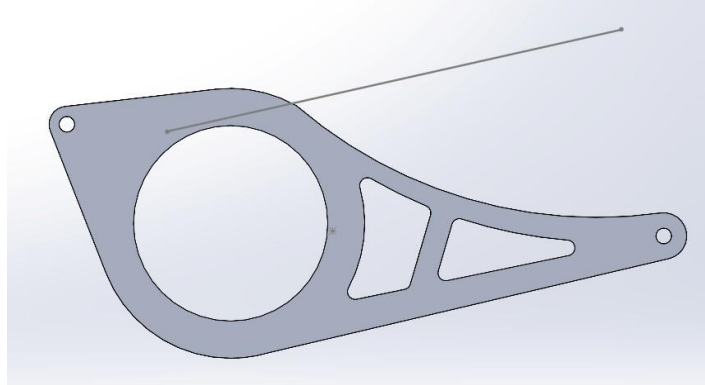


Figure 3: Left Differential Mount #2

Although this design was perfect for acting as rear impact protection for the Structural Equivalency Spreadsheet (SES), there were a lot of problems that came about as time went on, particularly that there was an issue that involved the position of the differential which caused the differential to move slightly forward. This means that design concept #2 had a lot of interference issues with the motor especially if the differential needed to be moved slightly up to install the chain. This led to the third and final design concept which involved moving the turnbuckles back to the bottom.

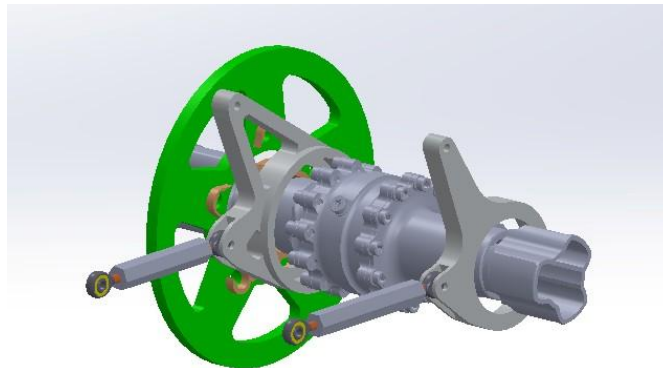


Figure 3: Drivetrain Assembly

This design involved mounting the mount from the top bar of the back of the frame which made it compliant with the regulations for rear impact protection (F.11.2.2 d). The left and right differential mounts have been designed differently because they experience differing amounts of load from the torque of the motor (the calculations of the load cases are below under the Calculations, FEA, and Modeling section). Since the left mount is closer to the chain, it experiences more force and therefore needed to have more material for a high stiffness-to-weight

ratio. The important design features of the mounts are the holes, extrusion above the flat surface, and the internal extrusion in the hole.

The diameter of the holes was determined by the press fit calculations shown in the Calculations section. Using the location interference calculations, the left mount has a hole of diameter 89.9715 mm while the right has a hole of diameter 79.9735 mm. The depth of the whole was determined by the thickness of the bearing (left: 18mm; right: 16mm). The thickness of the differential mounts was determined by the space that they would fit into the tabs on the frame. The slot made at the bottom of the mounts was made in order to fit the turnbuckles into the mount for the bolt of the nut. The right mount slot was made to be 30 deg to the plane formed by the mount when placed upright on a flat surface while the left mount slot is 45 deg to the flat surface of the mount.

Through FEA, the team was able to prove that this design can support differential, withstand loads from the motor, and with the turnbuckles at the bottom, they would be in compression and thus when extended the differential would move up more than half an inch, and allow for chain tensioning during installation and de-tensioning if there is a need to reduce the chain and there would be no interference issues with the motor mount or the half shafts interfering with the frame. It is also important to note that since the mounts are slightly different, the turnbuckles have different lengths and small nuts were added between the rod ends and the turnbuckles to prevent rotating when they need to be rigid. The length of the turnbuckles was determined such that the rod ends would be halfway inside when the chain is in tension. ...

In terms of material selection, 7075 T6 was used to make the differential mounts (this material has been used for the past 5 years for the differential mounts. This is because 7075 T6 has a high ultimate tensile strength of 510-540 MPa and a yield strength of at least 430-480 MPa which makes it very strong for the load cases from the motor. Furthermore aluminum is easy to manufacture from in-house.

The turnbuckles were made from low carbon steel which is very strong and will not buckle from the loads coming from the motor. However, this is very hard to deal with in terms of manufacturing in-house. It literally took two days to make the turnbuckles.

The hex bars were then fitted with JMX4 and JMXL4 rod ends on each side (right and left-handed). These are different directions so as to prevent the other rod end from moving when fastening the other one.

NFR22 had a total weight of 2.45 lbs. This year's design has a total weight of 1.29 lbs giving a total weight reduction of 1.16 lbs (47.3%).

EQ

Rear Impact Protection (EV Only)

F.11.2.2 From the rear, below 350mm, all HV components must be protected with an upper tube, a lower tube, and a diagonal tube or tubes completely triangulating the upper and lower tubes. Triangulation may be asymmetric.

If a plate replaces all three tubes, it must fully overlap the tractive side protection tubes.

F.11.2.2.a Increase from Size C to Size B if the accumulator is < 100mm (3.937in) from the rear tube.

F.11.2.2.b The Rear Protection must be fully triangulated back to the Upper and Lower SLS.

If a plate replaces all three tubes, 4x 30kN or 8x 15kN mounts are required.

Bolted joints must be documented if a removable panel or tube is used.

EQ					
F.11.2.2.a	Min distance from Accumulator to Rear Impact?	4.83	in		EQ
Tractive Rear Impact Protection		Minimum		Diff Carrier Diagonal	EQ
F.3.2.1.n	Example: 1.0in x 0.047in round	Size C	Round		EQ
F.3.2.1.c	Wall thickness:	0.047	0.065	in	EQ
	Outer Diameter (OD):	0.984	1	in	EQ
	Wall thickness:	1.2	1.7	mm	EQ
	Outer Diameter (OD):	25.0	25.4	mm	EQ
	Tube cross sectional area (A):	91	127	mm ²	EQ
	Tube second moment of inertia (I):	6695	8933	mm ⁴	EQ

F.11.2.2.b The entire top edge of the upper tube or plate must be at least 240mm above the lowest point of the top surface of the Lower SLS tube.

EQ		
Top surface of Lower SLS to top Rear Impact >=9.45in:	9.48	in

Outside rear differential carriers may be used for diagonal equivalence.

F.11.2.2.d The mounts or plate extend 25mm above and 25mm below the opening.

Holes in a plate must be too small for the differential to pass through.

The mounts are considered as beams, seen in the side view.

Sections should be taken in a plane parallel to the ground. *Plate minimums are more likely to be

Note: The minimum second moment and the minimum cross sectional area may not be in the same

Take the area and moment of inertia of the one mount.

If they are identical, these values can be multiplied by two.

If they are unique, add the values from the second mount.

If mounts are identical, R is the same centroid distance in each, and is the value used.

If mounts are asymmetric, average the centroid values for R.

EQ					
F.11.2.2.a	Rear Impact Tubes Replaced:	1			EQ
F.3.3-5	Material:	Steel	7075 Aluminum		EQ
F.3.4.2	Young's Modulus (E):	2.90E+07	1.04E+07	psi	EQ
	Yield Strength (Sy):	4.42E+04	7.32E+04	psi	EQ
	Ultimate Strength (Su):	5.29E+04	8.27E+04	psi	EQ
	Ultimate Is Critical Strength (S):	5.29E+04	8.27E+04	psi	EQ
	Differential carrier minimum cross sectional area (A):	1.41E-01	4.73E-01	in ²	EQ
	Differential carrier minimum second moment of inertia (I):	1.61E-02	1.13E+00	in ⁴	EQ
	Distance from longitudinal Edge to Moment of Inertia Centroid (R):	0.493	5.755	in	EQ
Elongation Modulus	$E_1 \cdot I_1 \leq E_2 \cdot I_2$	4.66E+05	1.18E+07	2519.4%	EQ
Ultimate Strength	$S_1 \cdot A_1 \leq S_2 \cdot A_2$	7.46E+03	3.91E+04	524.2%	EQ
Bending	$4 \cdot S_1 \cdot I_1 / r \leq 4 \cdot S_2 \cdot I_2 / r$	6.90E+03	6.50E+04	940.8%	EQ
Deflection	Bending $1 / (48 \cdot EI)$	3.08E-04	1.22E-05	4.0%	EQ
Energy	$0.5 \cdot \text{Bending}^2 / (48 \cdot EI)$	1.06E+00	3.74E+00	351.3%	EQ

Interactions:

Frame

The frame was made by Drew Hill and Kayleigh Peng and is the main chassis component that practically holds the car together. The team worked with Drew and Kayleigh, especially on the discussion of the position of the tabs on the frame and whether that was compliant with the regulations.

Sprocket Adapter

The sprocket adapter was made by Stella Heo and Nathan Lee, who are a part of the Suspension subteam and was manufactured to connect the sprocket to the drivetrain and differential. The team worked with Stella and Nathan to check for any interference issues that the differential mounts might have with the sprocket adapter and to determine the torque that comes from the motor.

Undertray

The undertray was designed and manufactured by Peter Akcan. The undertray generates downforce by creating a low-pressure region below the car. The team worked with Peter to determine if the differential mounts to not have any interference issues with the undertray.

Motor Mount

The motor mount was designed and manufactured by Armaan Jain and Drake Vogelpohl, who are a part of the Suspension subteam. The motor mount was designed to support the motor from its position onto the frame. The team worked with Armaan and Drake to determine if the differential will not interfere with the motor mount when moving it up to install the chain.

Calculations, FEA, and Modeling:

The calculations and simulations of the differential mounts have been done on the basis of treating the differential with the mounts as bar carrying forces generated from the torque of the chain.

Left Differential Mount Topology Optimization:

Fixtures:

- Fixed Hinge at the top tab hole Loads:
- Bearing Loads, Coordinate System 3¹, 9987N

¹ This is a reference geometry created at the center of the bearing hole of the mount.

- Remote Displacement¹, Distributed, Coordinate System 1³, Translational
Components: 0 mm, 0 mm, 0 mm, Rotational components: -, 0 deg, 0 deg,
Reference coordinates: (0,0,0)mm Goals and Constraints:
 - Minimize mass
- Manufacturing Controls:
- Preserved region, 7 faces (the tab holes and differential holes),
Preserved Area Depth: 0.25in Mesh:
 - Mesh type: Solid Mesh
 - Mesher Used: Blended curvature-based mesh
 - Jacobian points for High quality mesh: 16 points
 - Max Element Size: 0.179372 in
 - Min Element Size: 0.101054 in
 - Mesh quality: High
 - Total nodes: 22655
 - Total elements: 13428
 - Maximum Aspect Ratio: 29.651
 - Percentage of elements\with Aspect Ratio < 3: 99.1
 - Percentage of elements\with Aspect Ratio > 10: 0.194

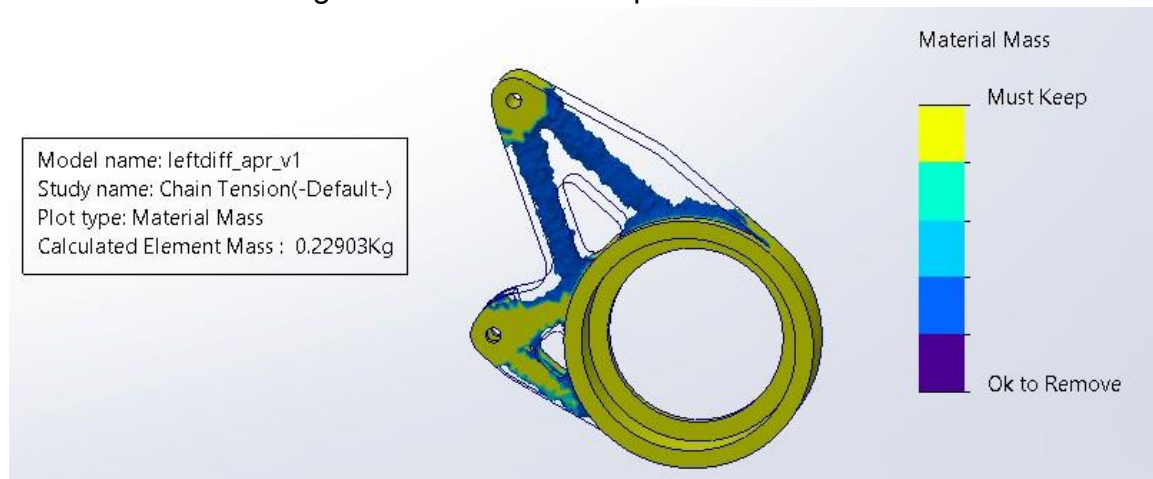


Figure 4: Left Differential Mount Topology Optimization

Right Differential Mount Topology Optimization:

Fixtures:

- Fixed Hinge at the top tab hole Loads:
- Bearing Loads, Coordinate System 3, -1730N

¹ This is to constrain the mount to one-dimensional movement when bolted onto the frame.

³ The coordinate system was coincident with the center of the tabs on the frame

- Remote Displacement, Distributed, Coordinate System 1, Translational

Components: 0 mm, 0 mm, 0 mm, Rotational components: -, 0 deg, 0 deg,
Reference coordinates: (0,0,0)mm Goals and Constraints:

- Minimize mass

Manufacturing Controls:

- Preserved region, 7 faces (the tab holes and differential holes), Preserved Area Depth: 0.25in Mesh:

- Mesh type: Solid Mesh
- Mesher Used: Blended curvature-based mesh
- Jacobian points for High quality mesh: 16 points
- Max Element Size: 0.179372 in
- Min Element Size: 0.101054 in
- Mesh quality: High
- Total nodes: 22655
- Total elements: 13428
- Maximum Aspect Ratio: 29.651
- Percentage of elements\with Aspect Ratio < 3: 99.1
- Percentage of elements\with Aspect Ratio > 10: 0.194

Unfortunately I lost the pictures to the topology optimization study. However, running this study with the part may take a long time but will give you the results that led to final design.

Press Fit Calculations:

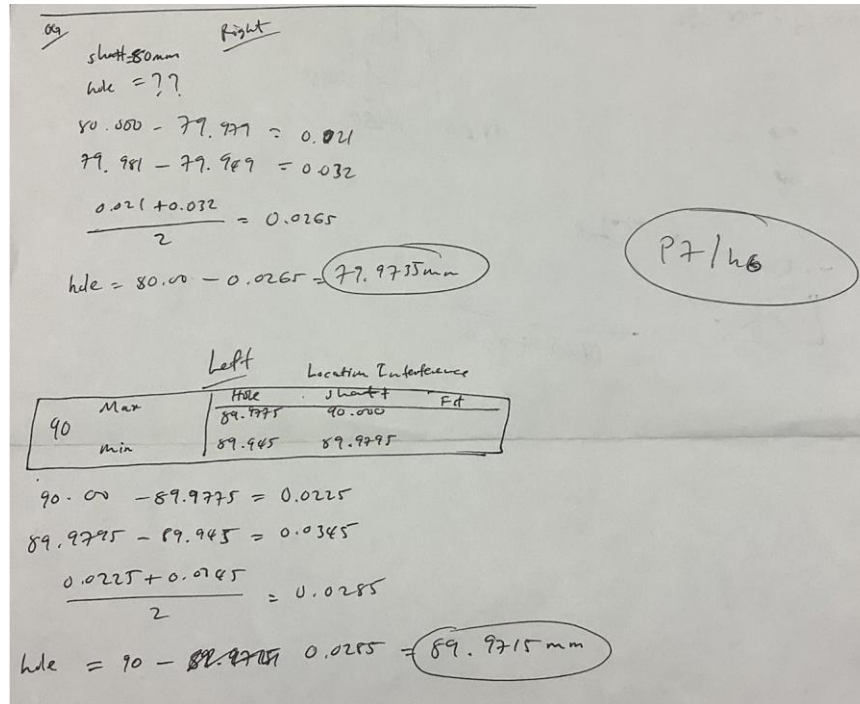


Figure 5: Press fit Calculations

Bearing Load Calculations:

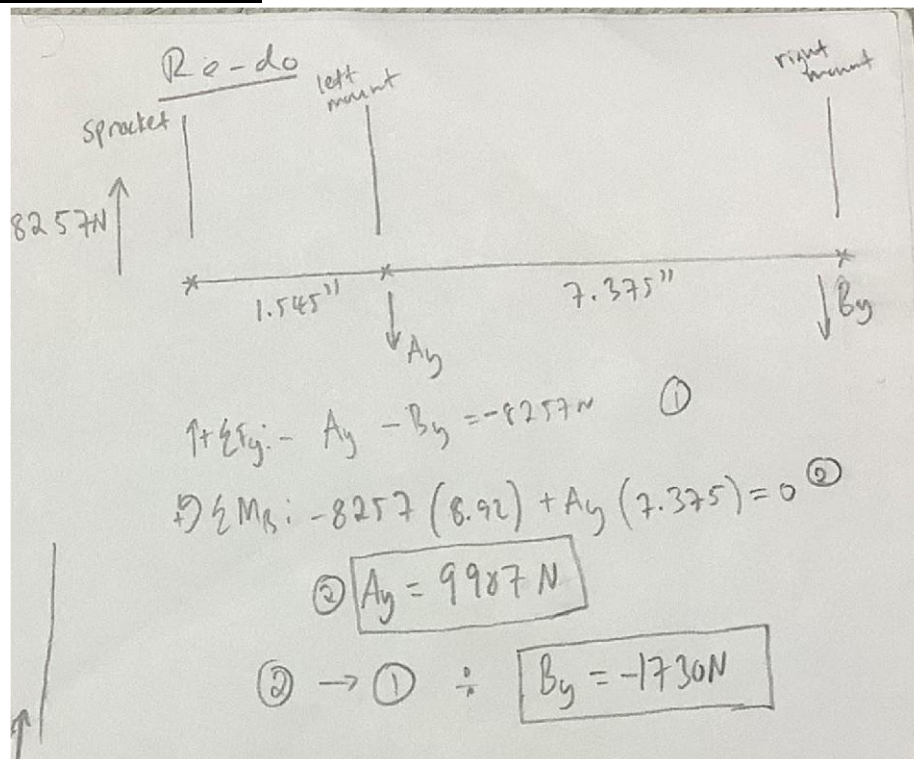


Figure 6: Bearing Load Calculations

The force from the tension was determined by dividing the maximum torque from the motor by the radius of the sprocket. The radius of the sprocket was calculated by taking half of the pitch radius.

Left Mount - Upward Bearing Load Fixtures:

- Fixed Hinge at the top tab hole Loads:
- Bearing Loads, Coordinate System 3, 9987N
- Remote Displacement, Distributed, Coordinate System 1,

Translational

Components: 0 mm, 0 mm, 0 mm, Rotational components: -, 0 deg, 0 deg,
Reference coordinates: (0,0,0)mm Mesh:

- Mesh type: Solid Mesh
- Mesher Used: Blended curvature-based mesh
- Jacobian points for High quality mesh: 16 points
- Max Element Size: 0.0938167 in
- Min Element Size: 0.0312719 in
- Mesh quality: High
- Total nodes: 108679
- Total elements: 70186

Model name: leftdiff_apr_v1
Study name: Tensional Forces(-Default-)
Plot type: Static nodal stress Stress1

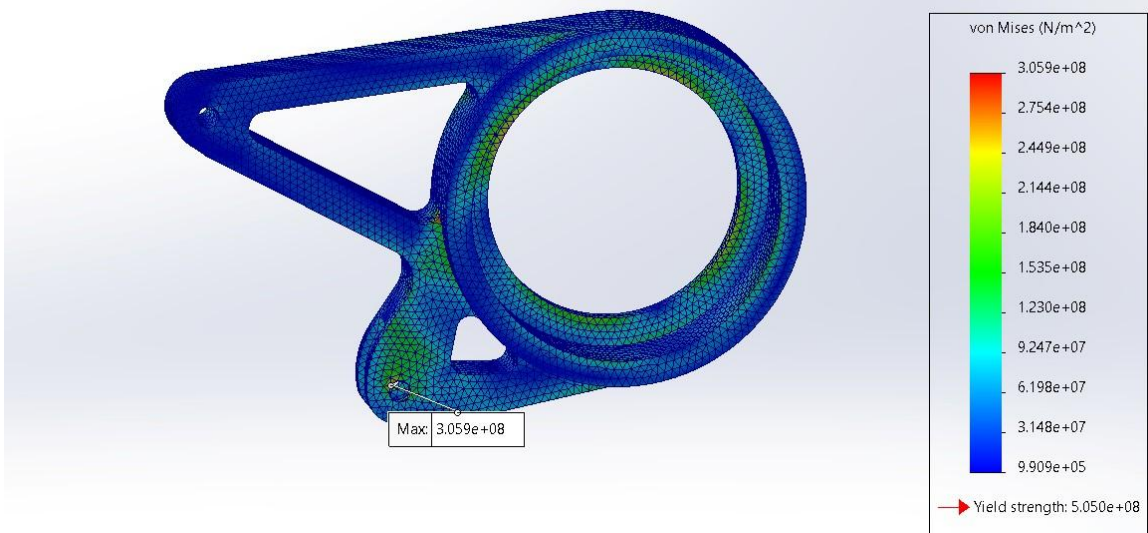


Figure 7: Stress Plot for the Left Differential Mount

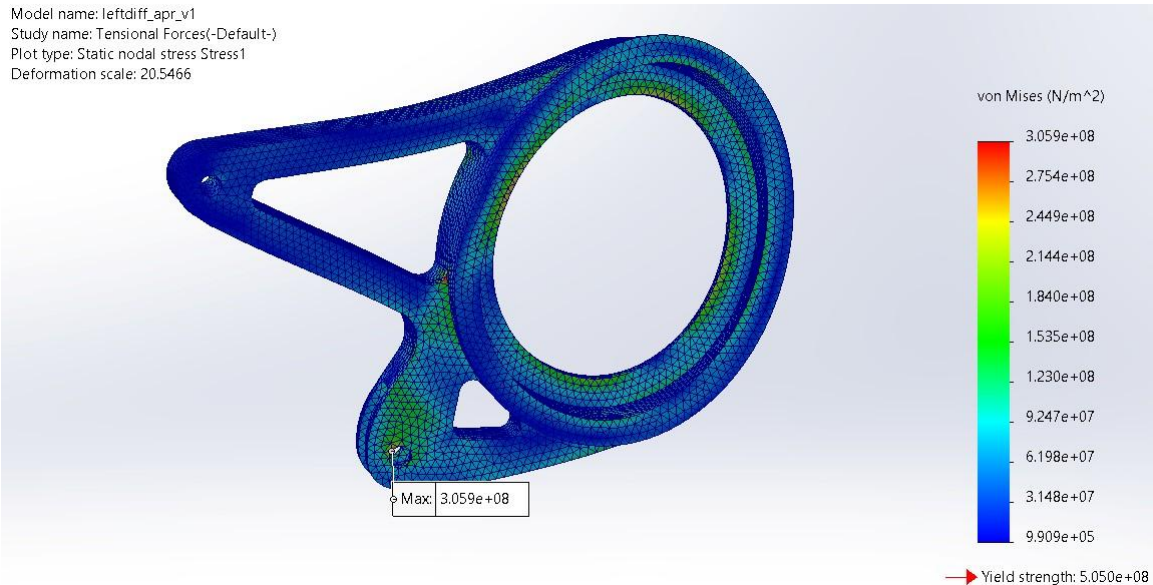


Figure 8: Deformation Plot for the Left Differential Mount

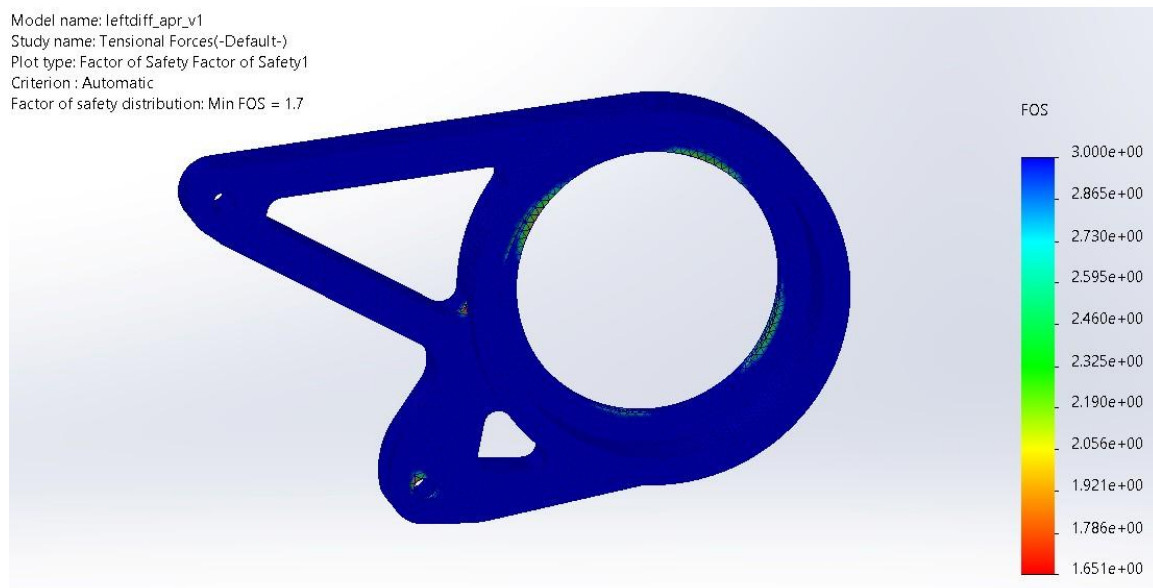


Figure 9: Factor of Safety Plot for the Left Differential Mount

Right Mount - Downward Bearing Load Fixtures:

- Fixed Hinge at the top tab hole Loads:
- Bearing Loads, Coordinate System 3, -1730N
- Remote Displacement, Distributed, Coordinate System 1,
Translational Components: 0 mm, 0 mm, 0 mm, Rotational components: -, 0
deg, 0 deg, Reference coordinates: (0,0,0)mm Mesh:
- Mesh type: Solid Mesh
- Mesher Used: Blended curvature-based mesh

- Jacobian points for High quality mesh: 16 points
- Max Element Size: 0.0855652 in
- Min Element Size: 0.0285214 in
- Mesh quality: High
- Total nodes: 96160
- Total elements: 62414

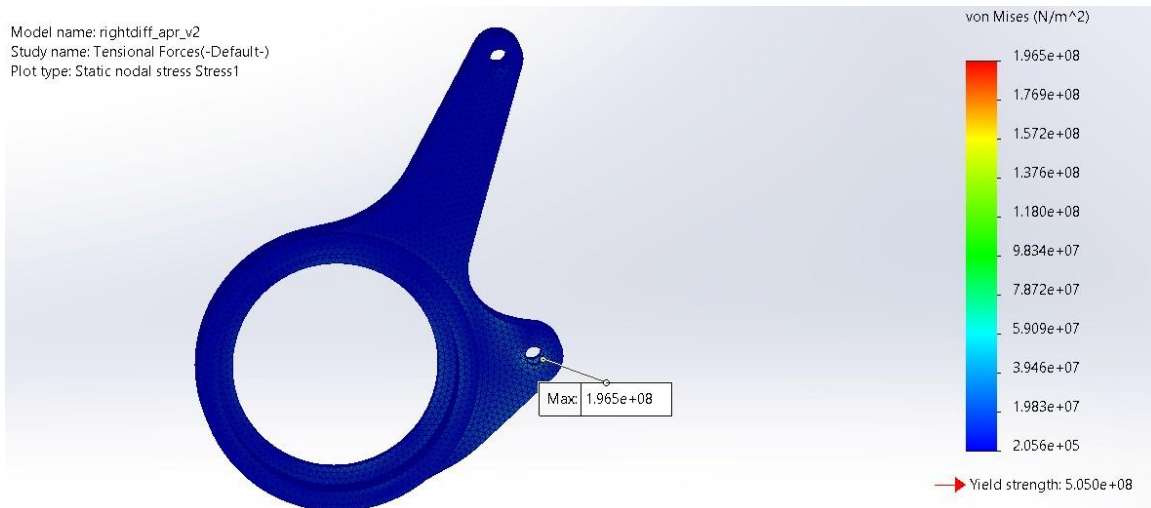


Figure 9: Stress Plot for the Right Differential Mount

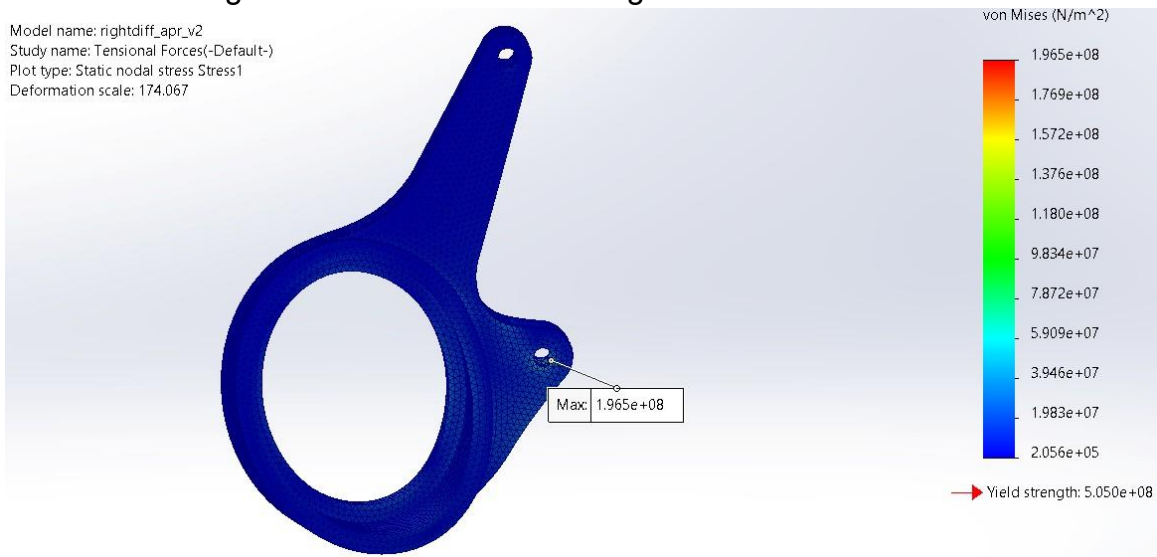


Figure 10: Deformation Plot for the Right Differential Mount

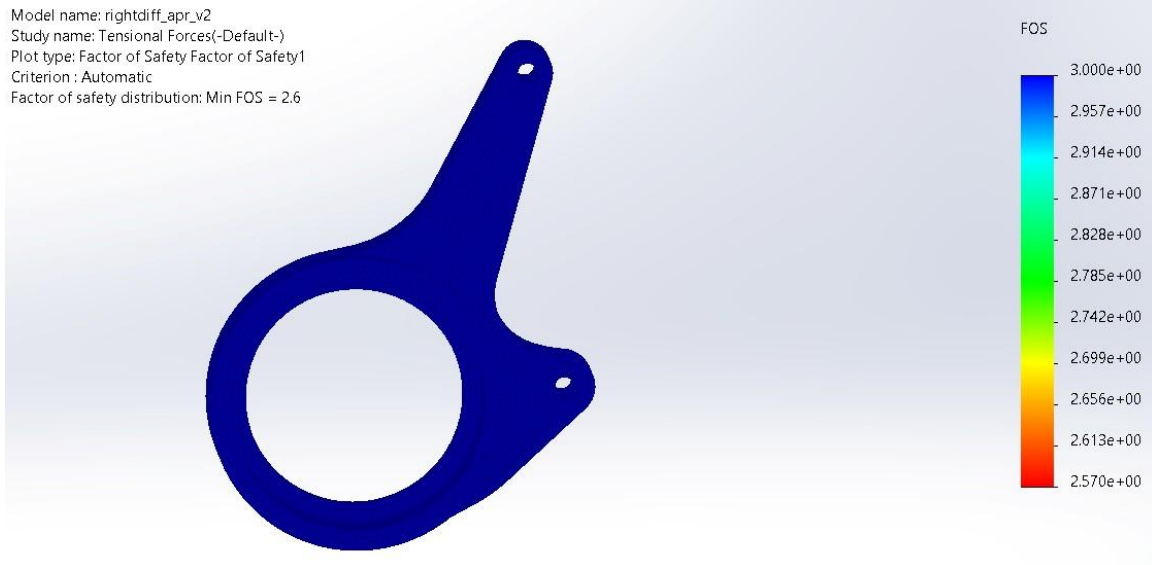


Figure 11: Factor of Safety Plot for the Right Differential Mount

Manufacturing Plan:

Differential Mounts

The two differential mounts were manufactured in-house from the same blank with the dimensions: 13" x 9" x 1.5", using the Haas CNC machine. The material of the blank was 7075-T6 as mentioned before. Two major operations were done on the blank on the CNC Mill. The first operation was to face off the blank. The blank was then flipped upside down to do the second operation. The CAM for the mounts was done using Fusion 360.

Here is the order of sub-operations in the second operation:

1. Face off till the top surface of the left mount (3 in Face Mill)
2. 2D Adaptive to make the bearing holes ($\frac{3}{4}$ in Flat Indexable End Mill)
3. 2D Adaptive to plunge through the smaller hole to fully form bearing housing ($\frac{3}{4}$ in Flat Indexable End Mill)
4. 2D Contour to remove more material in the bearing hole (12 mm Flat End Mill)
5. 2D Contour to remove more material in the smaller hole for bearing (12 mm Flat End Mill)
6. 2D Contour to remove material around the top extrusion of the right differential mount (12 mm Flat End Mill)
7. 2D Adaptive to remove material on the lower surface for the right mount (12 mm Flat End Mill)
8. 2D Adaptive to make the holes made to reduce weight in the left mount ($\frac{1}{4}$ in Flat End Mill)
9. 2D Adaptive to remove material on the lower surface for the left mount ($\frac{1}{4}$ in Flat End Mill)
10. 2D Contour to remove material around the top surface of the left mount ($\frac{1}{4}$ in Flat End Mill)
11. Center Drill the tab and turnbuckle holes in the left mount ($\frac{1}{8}$ in Center Drill)
12. Center Drill the tab and turnbuckle holes in the right mount ($\frac{1}{8}$ in Center Drill)
13. Drill all the holes ($\frac{1}{4}$ in Drill)
14. 2D Contour to cut out the right mount ($\frac{1}{4}$ in Flat End Mill)
15. 2D Contour to cut out the left mount ($\frac{1}{4}$ in Flat End Mill)

The mounts were then removed from the remaining blank material using the vertical bandsaw. The CNC operations were then followed by operations on the manual mill to make the slots. 30 deg and 45 deg angle blocks were used to fixture the mounts on the mill. A $\frac{5}{8}$ in end mill was then just passed straight through on the x-direction to make the slots.

Turnbuckles

The turnbuckles were made from a low carbon steel 2 ft long hex bar. The hex bar was divided up using the horizontal band saw and then cut to the proper length using the lathe. Each turnbuckle was drilled an inch on each side using a 0.213 in drill bit (center drilling was done before drilling). One side on each turnbuckle was threaded using a right-hand 1/4-28 tap while the other side was tapered using a left-hand 1/4-28 tap.

Assembly

The assembly of each differential mount started by press fitting the bearings into the mounts using the manual arbor press.

Parts list:

- Deep groove ball bearing, SKF 6011-2z
- Radial ball bearing, SKF 6010-2z
- 1/4 in Bolts
- 1/4 in Nuts
- Steel Hex Bar, 6607K35
- JMX4 rod ends
- JMXL4 rod ends
- 7075-T6 blank

Pictures:



Figure 11: Final Manufactured Part



Figure 12: Differential Mounts and turnbuckles
assembled in tabs on frame

During the manufacturing process, the integrity of the designs were maintained. The differential mounts were manufactured via Haas CNC, so the CAM that was imported into the CNC machine came straight from the CAD. However, the press fit process to put the bearings on the differential changed. Originally, the plan was to heat the bearings, expand them, and fit them onto the differential. Then, the bearings would cool and tightly fasten onto the differential. After the bearings were secure, the differential mounts would be fastened to the bearings. Instead, the bearings were placed in the differential mounts first. They were manually pressed into the mounts with the arbor press. Then, the mounts plus bearings were manually pressed to the differential with the arbor press. There was also an alteration when fixing the differential mounts to the tabs. The width of the differential mounts were slightly larger than the width of the tab openings for the top tabs. To make the mounts fit inside the tabs, the sides of the tabs were grinded out to create more room.

The most difficult part of creating the differential mounts is the design process. The differential mounts must be able to rotate from the position of the turnbuckle around where it is fixed to the frame. The turnbuckle must be angled so that when it is lengthened or shortened, the differential mount moves in the direction of the motor sprocket so that the chain is can be properly tensioned and detensioned. After multiple redesigns, the best way to achieve this is to place the turnbuckles on the bottom of the frame and fix the differential mounts to the top of the frame. This way, lengthening the turnbuckles will move the differential closer to the motor, detensioning the chain. And fastening the turnbuckles will have the opposite effect.

The last big concern is that when designing the mounts, ensure that there is enough clearance between the mounts and frames so that when you tighten and loosen the turnbuckles, the mounts are able to move freely without hitting the frame or other components of the car.

Other points:

- Although the differential mounts did not have any rules, READ THE WHOLE RULE BOOK!!!!
- Talk to past differential mount design engineers: Siby Suriyan, Danish Galebotswe, Marco Wang, Tommy Li
- Read past documentation and have a scope meeting with your lead and chief engineer before starting your design process.
- Follow the timeline provided by your lead. If the PDR is scheduled for October, aim for October.
- Don't be afraid to ask questions if you do not understand the functions of the differential mounts.

- Good luck!!!